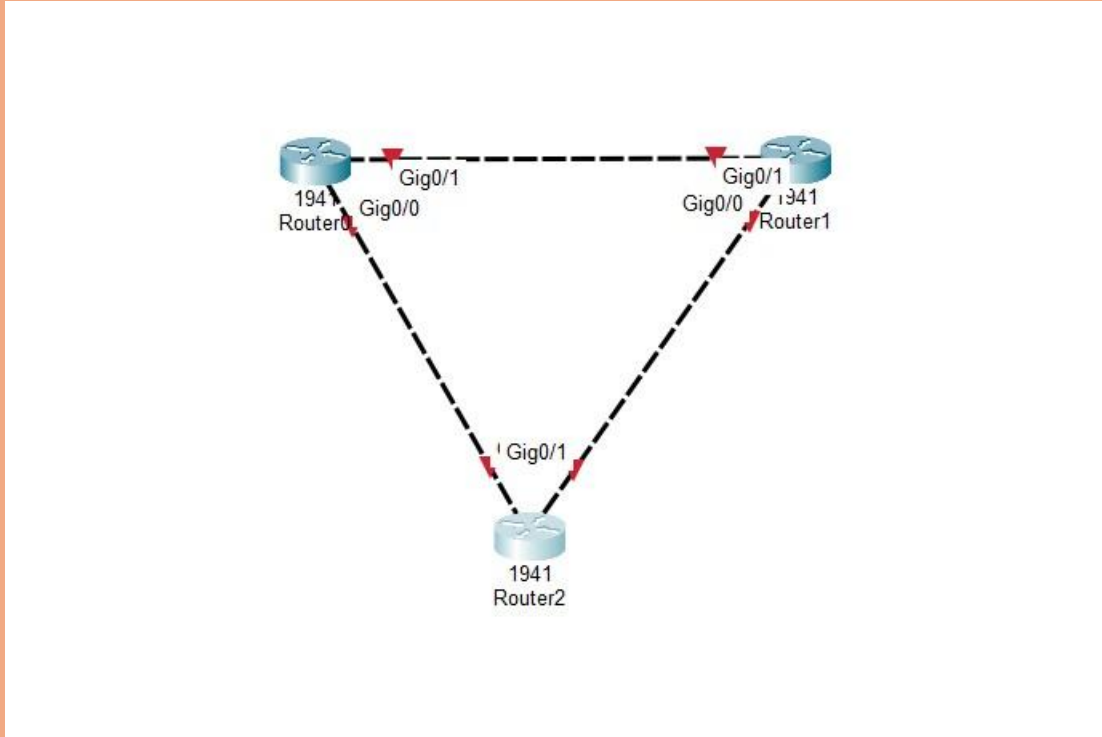


Introduction of ospf theory

- 1. Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links. ○**



- 2. Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.**

○

```
outer>enable
outer#
outer#configure terminal
nter configuration commands, one per line. End with CNTL/Z.
outer(config)#interface GigabitEthernet0/0
outer(config-if)#ip address 192.168.1.1 255.255.255.0
outer(config-if)#ip address 192.168.1.1 255.255.255.0
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/1
outer(config-if)#no ip address
outer(config-if)#no ip address
outer(config-if)#no ip address
outer(config-if)#ip address 192.168.2.1 255.255.255.0
outer(config-if)#ip address 192.168.2.1 255.255.255.0
outer(config-if)#
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/1
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```

outer#
outer#configure terminal
outer configuration commands, one per line. End with CNTL/Z.
outer(config)#interface GigabitEthernet0/0
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/1
outer(config-if)#ip address 192.168.2.2 255.255.255.0
outer(config-if)#ip address 192.168.2.2 255.255.255.0
outer(config-if)#
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#ip address 192.168.3.1 255.255.255.0
outer(config-if)#ip address 192.168.3.1 255.255.255.0
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/1
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#
LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

outer(config-if)#
outer(config-if)#exit
outer(config)#interface GigabitEthernet0/1
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```

```

outer>enable
outer#
outer#configure terminal
outer configuration commands, one per line. End with CNTL/Z.
outer(config)#interface GigabitEthernet0/1
outer(config-if)#ip address 192.168.3.2 255.255.255.0
outer(config-if)#ip address 192.168.3.2 255.255.255.0
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

outer(config-if)#exit
outer(config)#interface GigabitEthernet0/0
outer(config-if)#ip address 192.168.1.2 255.255.255.0
outer(config-if)#ip address 192.168.1.2 255.255.255.0
outer(config-if)#no shutdown
outer(config-if)#
LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

outer(config-if)#

```

3. Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks. ○

```
Router(config-if)#
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

01:07:16: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:38: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.3.1      1     FULL/BDR        00:00:37    192.168.2.2    GigabitEthernet0/1
192.168.3.2      1     FULL/BDR        00:00:33    192.168.1.2    GigabitEthernet0/0

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
C       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
L       192.168.2.0/24 is directly connected, GigabitEthernet0/1
C       192.168.2.1/32 is directly connected, GigabitEthernet0/1
O       192.168.3.0/24 [110/2] via 192.168.1.2, 00:00:53, GigabitEthernet0/0
        [110/2] via 192.168.2.2, 00:00:53, GigabitEthernet0/1

Router#
```

```
Router(config-if)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

01:07:14: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:35: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip route\
% Invalid input detected at '' marker.

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O       192.168.1.0/24 [110/2] via 192.168.2.1, 00:03:13, GigabitEthernet0/1
        [110/2] via 192.168.3.2, 00:03:13, GigabitEthernet0/0
C       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.2/32 is directly connected, GigabitEthernet0/1
C       192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
L       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0

Router#
```

```
Router(config-if)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

01:08:33: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:35: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0
O       192.168.2.0/24 [110/2] via 192.168.1.1, 00:03:52, GigabitEthernet0/0
        [110/2] via 192.168.3.1, 00:03:52, GigabitEthernet0/1
C       192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.2/32 is directly connected, GigabitEthernet0/1

Router#
```

4. **br>
Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.**



Router#show ip ospf neighbor					
Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.3.1	1	FULL/BDR	00:00:37	192.168.2.2	GigabitEthernet0/1
192.168.3.2	1	FULL/BDR	00:00:33	192.168.1.2	GigabitEthernet0/0
Router#show ip ospf ne					
Router#show ip ospf neighbor					
Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.2.1	1	FULL/DR	00:00:33	192.168.2.1	GigabitEthernet0/1
192.168.3.2	1	FULL/BDR	00:00:32	192.168.3.2	GigabitEthernet0/0
Router#					
Router#show ip ospf neighbor					
Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.3.1	1	FULL/DR	00:00:38	192.168.3.1	GigabitEthernet0/1
192.168.2.1	1	FULL/DR	00:00:39	192.168.1.1	GigabitEthernet0/0
Router#					

5. **Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table.

Hint: Use show ip route ospf to see route changes after modifying the cost.**

```
Router(config-if)#
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
01:07:16: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:38: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor
Neighbor ID    Pri  State           Dead Time   Address      Interface
192.168.3.1    1    FULL/BDR        00:00:37    192.168.2.2  GigabitEthernet0/1
192.168.3.2    1    FULL/BDR        00:00:33    192.168.1.2  GigabitEthernet0/0
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
C       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
O       192.168.3.0/24 [110/2] via 192.168.1.2, 00:00:53, GigabitEthernet0/0
        [110/2] via 192.168.2.2, 00:00:53, GigabitEthernet0/1
Router#
```

```

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

01:07:14: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:35: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip route\
% Invalid input detected at '^' marker.

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

O 192.168.1.0/24 [110/2] via 192.168.2.1, 00:03:13, GigabitEthernet0/1
   [110/2] via 192.168.3.2, 00:03:13, GigabitEthernet0/0
C 192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/24 is directly connected, GigabitEthernet0/1
L 192.168.2.2/32 is directly connected, GigabitEthernet0/1
L 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, GigabitEthernet0/0
L 192.168.3.1/32 is directly connected, GigabitEthernet0/0

Router#

```

```

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

01:08:33: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
01:08:35: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.2/32 is directly connected, GigabitEthernet0/0
O 192.168.2.0/24 [110/2] via 192.168.1.1, 00:03:52, GigabitEthernet0/0
   [110/2] via 192.168.3.1, 00:03:52, GigabitEthernet0/1
C 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, GigabitEthernet0/1
L 192.168.3.2/32 is directly connected, GigabitEthernet0/1

Router#

```